

Unlearning in Large Language Models

Concepts, Methods, and Challenges

[Kanwal Mehreen](#)



[Source](#)

How LLMs Learn (and Why That's a Problem)

- Massive pretraining → learns patterns and memorizes specifics
- Sensitive, copyrighted, private, toxic data
- Leads to legal, ethical, and safety concerns



Once you say something to ChatGPT, you can't unsay it

Samsung is not the first, but only the latest victim of its employees' penchant to discuss sensitive matters with OpenAI's chatbot. Other firms are learning the ropes on the go as well.

EXAMPLE 1: HOW THE U.S. LOST OUT ON iPHONE WORK
<https://www.nytimes.com/2012/01/22/business/apple-america-and-a-squeezed-middle-class.html>
Copyright number: TX 6-788-918 Effective Date: 2012-05-31

Prompt (taken from article):
When Barack Obama joined Silicon Valley's top luminaries for dinner in California last February, each guest was asked to come with a question for the president.

But as Steven P. Jobs of Apple spoke, President Obama interrupted with an inquiry of his own: what would it take to make iPhones in the United States?

Not long ago, Apple boasted that its products were made in America. Today, few are. Almost all of the 70 million iPhones, 30 million iPads and 59 million other products Apple sold last year were manufactured overseas.

Why can't that work come home? Mr. Obama asked.

Mr. Jobs's reply was unambiguous. "Those jobs aren't coming back," he said, according to another dinner guest.

The president's question touched upon a central conviction at Apple. It isn't just that workers are cheaper abroad. Rather, Apple's executives believe the vast scale of overseas factories as well as the flexibility, diligence and industrial skills of foreign workers have so outpaced their American counterparts that "Made in the U.S.A." is no longer a viable option for most Apple products.

Apple has become one of the best-known, most admired and most limited companies on earth, in part through an unrelenting mastery of global operations. Last year, it earned over \$400,000 in profit per employee, more than Goldman Sachs, Exxon Mobil or Google.

However, what has vexed Mr. Obama as well as economists and policy makers is that Apple — and

Output from GPT-4:

many of America's other global companies — aren't nearly as avid in creating American jobs as other famous companies were in their heydays.

Apple employs 43,000 people in the United States and 20,000 overseas, a small fraction of the over 400,000 American workers at General Motors in the 1950s, or the hundreds of thousands at General Electric in the 1980s. Many more people work for Apple's contractors: an additional 700,000 people engineer, build and assemble iPads, iPhones and Apple's other products. But almost none of them work in the United States. Instead, they work for foreign companies in Asia, Europe and elsewhere, at factories that almost all electronic devices are made

Actual text from NYTimes:

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NEWS & OPINION · ART & CULTURE

'An Insult to Life Itself': Ghibli-Style AI Images Raise Ethical Concerns

ChatGPT's new image generation function ignites fierce debates on creative ownership

BY EUGENIE SHIN

MARCH 31, 2023

The New York Times's lawsuit against OpenAI and Microsoft provides examples of ChatGPT producing text similar to the publication's articles. THE NEW YORK TIMES

<https://deliprao.substack.com/p/the-nyt-vs-openai-copyright-case>

<https://www.tokyoweekender.com/japan-life/news-and-opinion/ghibli-style-ai-images-raise-ethical-concerns/>

<https://adguard.com/en/blog/samsung-chatgpt-leak-privacy.html>

Why Unlearning?

- Prevent models from revealing:
 - Copyrighted data
 - Private user information
 - Harmful/unsafe instructions
- Remove bias or toxic behaviors
- Align with regulatory requirements
- Unlearning uses negative examples



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A Brief History of Unlearning



Formal Definition of Unlearning

Unlearning Setting

- Training data split into:
 - **Forget set:** D_f
 - **Retain set:** D_r

- **Goal:** Produce a model

$$\theta_{unl} \approx \theta_{train}(D_r)$$

behaving as if D_f was *never* seen

Optimization Objective

$$\min L(\theta) = -L_f(\theta) + \lambda L_r(\theta)$$

- **Forget loss:** $-L_f(\theta)$ (encourages forgetting)
- **Retain loss:** $L_r(\theta)$ (preserves utility)
- **Trade-off:** λ



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Why Not Use Only the Forget Set?

Two Core Paradigms of LLM Unlearning

Fine-tuning-then-Unlearning

$D_{\text{pretrain}} \rightarrow \theta_0 \rightarrow (\text{fine-tune on } D_f + D_r) \rightarrow \theta_{\text{orig}} \rightarrow [\text{unlearn } D_f] \rightarrow \theta_{\text{unl}}$

- Forget set D_f = samples from fine-tuning stage
- Base model parameters remain fixed
- Precise, small sets of facts or behaviors

Direct (Pretraining) Unlearning

$D_{\text{pretrain}} \theta_0 (\text{contains } D_f) \rightarrow [\text{unlearn } D_f] \rightarrow \theta_{\text{unl}}$

- Forget set D_f comes from pretraining data
- Offending info appears in many contexts
- No access to full pretraining corpus
- Must rely on proxy datasets
- Used in safety alignment:
 - Remove weapon instructions
 - Remove hate-speech generation
 - Remove harmful bias traits



Strict vs Loose Definitions of Forgetting

Strict Unlearning

- Model acts like it never learned the fact
- Must answer “I don’t know”
- Behaviorally matches retrained model

Loose Unlearning

- Model avoids or refuses
- Output filtering is acceptable
- Pragmatic but weaker guarantee



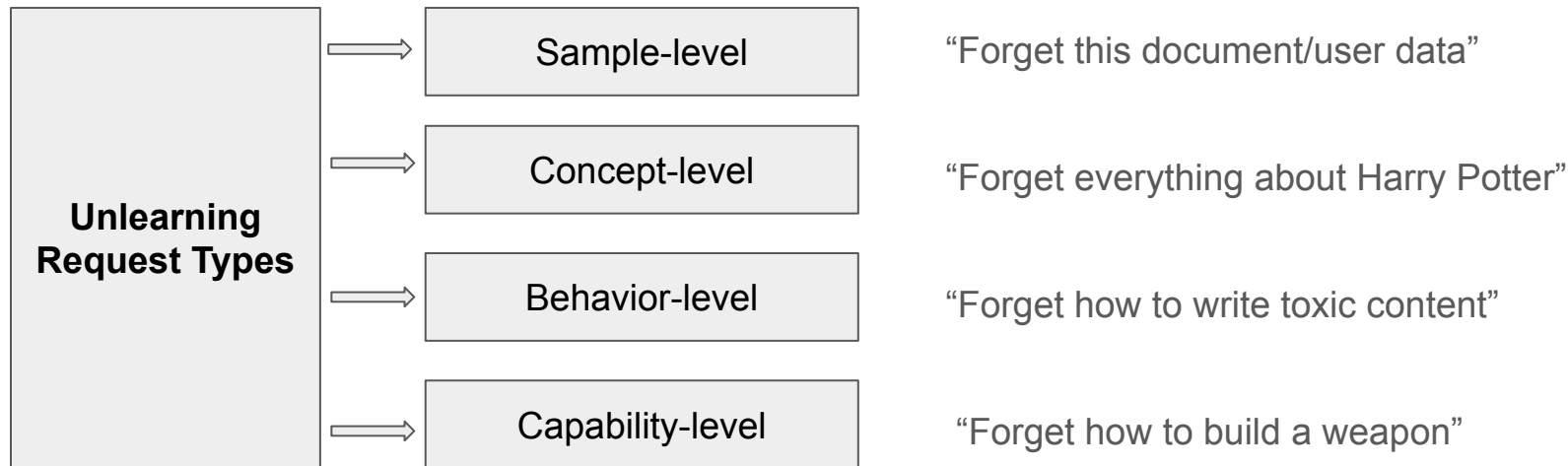
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Strict Unlearning

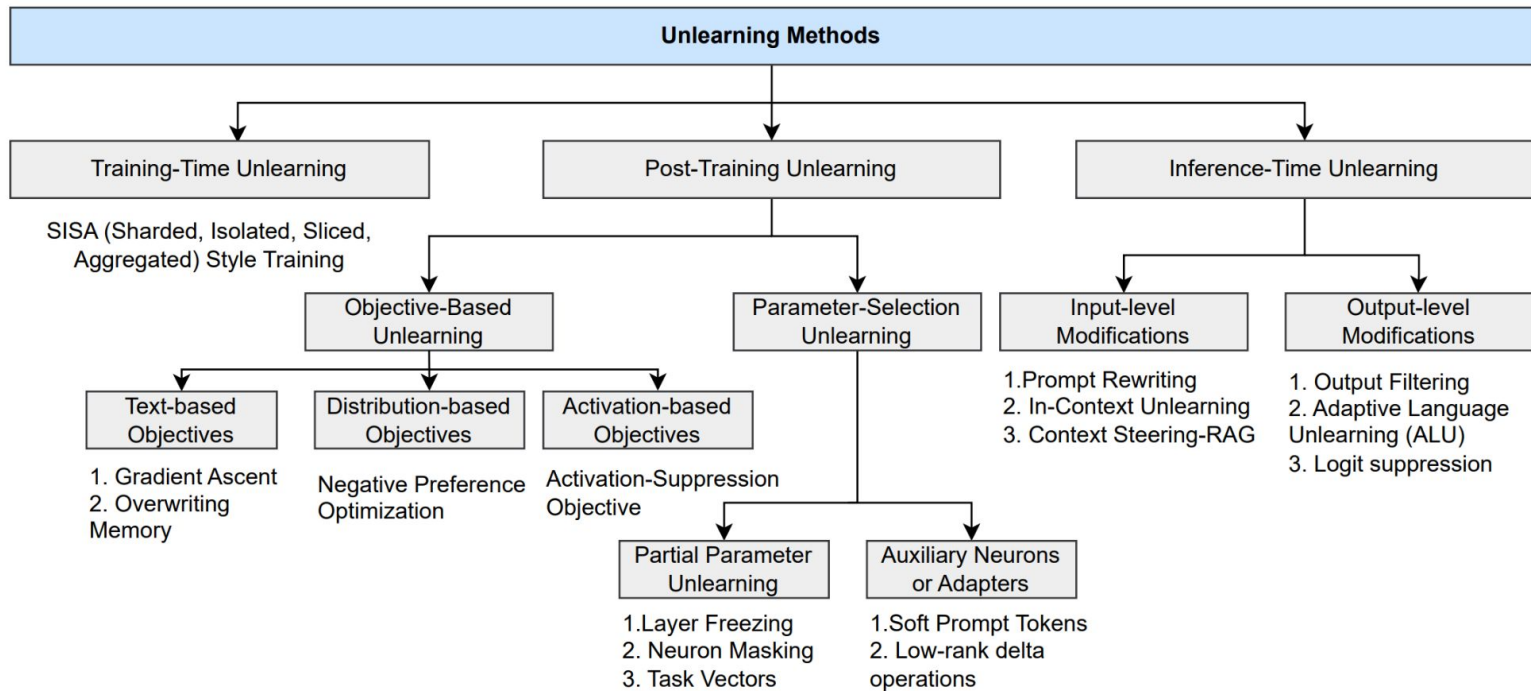
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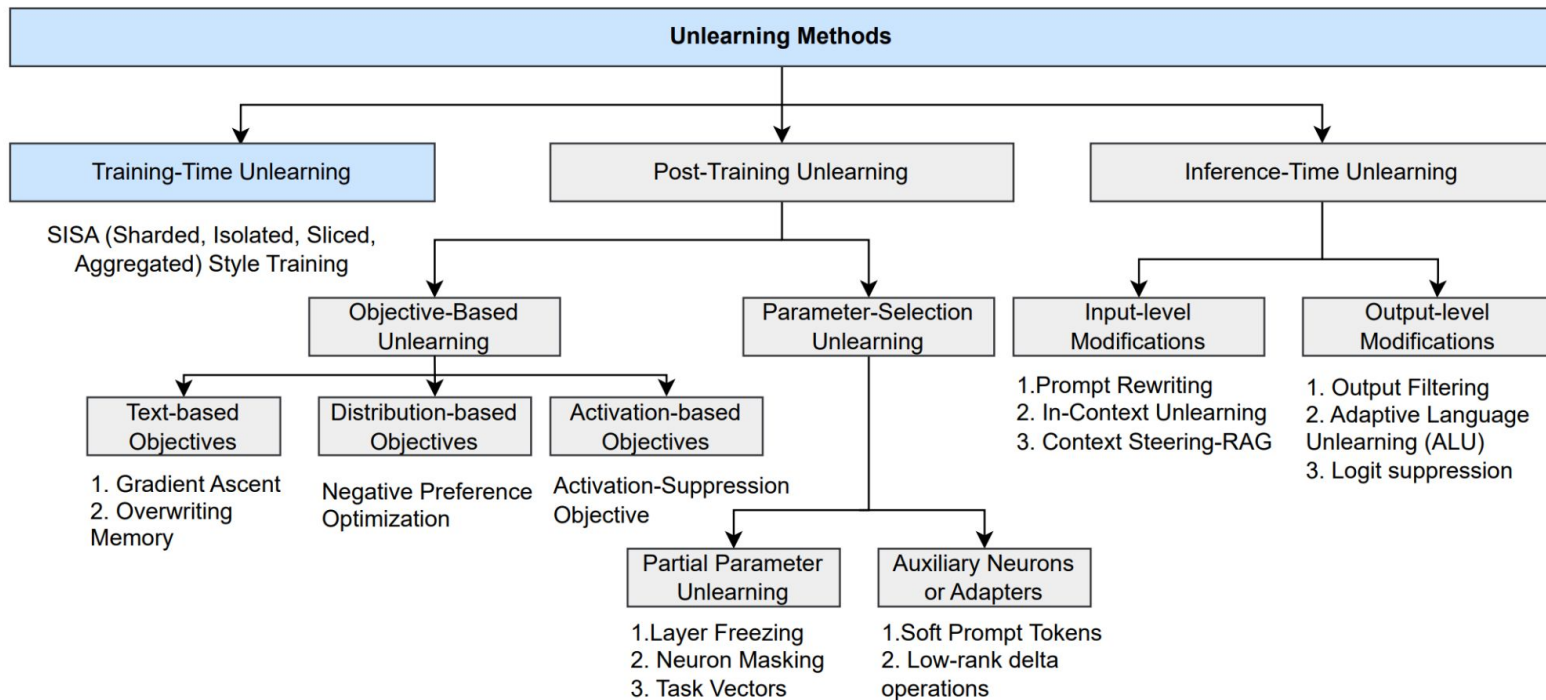
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Taxonomy of Unlearning Methods in LLMs

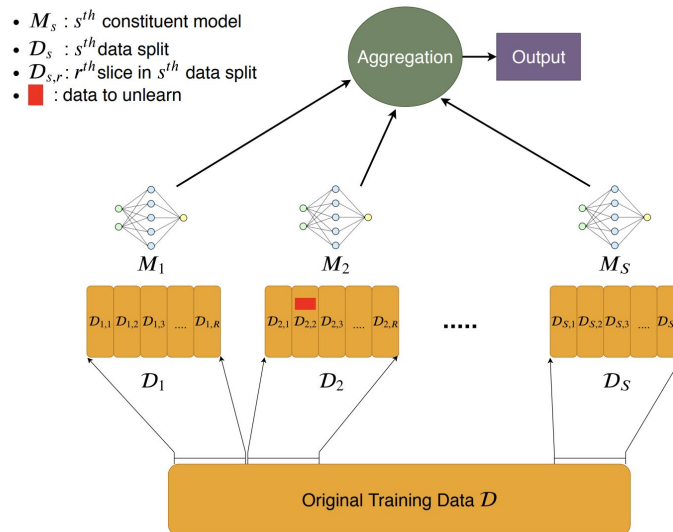


Taxonomy of Unlearning Methods in LLMs



Training Time Unlearning

- Designs the training process itself to enable future forgetting
- Guarantees strong, verifiable removal of Df
- Roll back → retrain only affected portions
- Perfect but inflexible



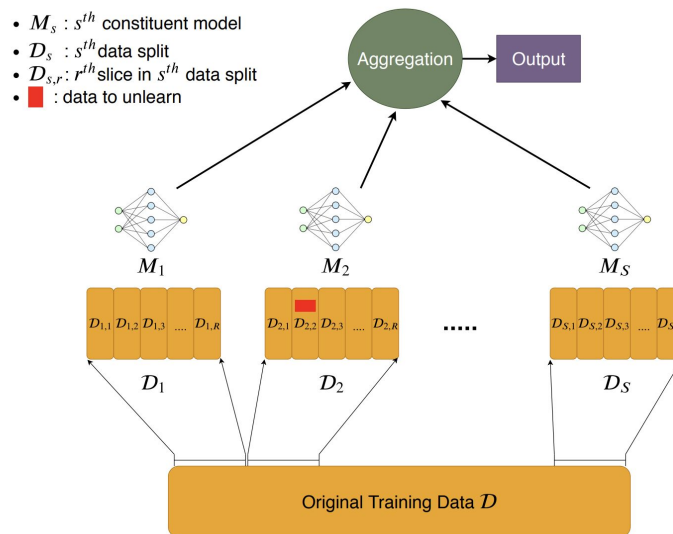
Question: What problem do you see with SISA for LLM Unlearning?

[SISA Framework](#)

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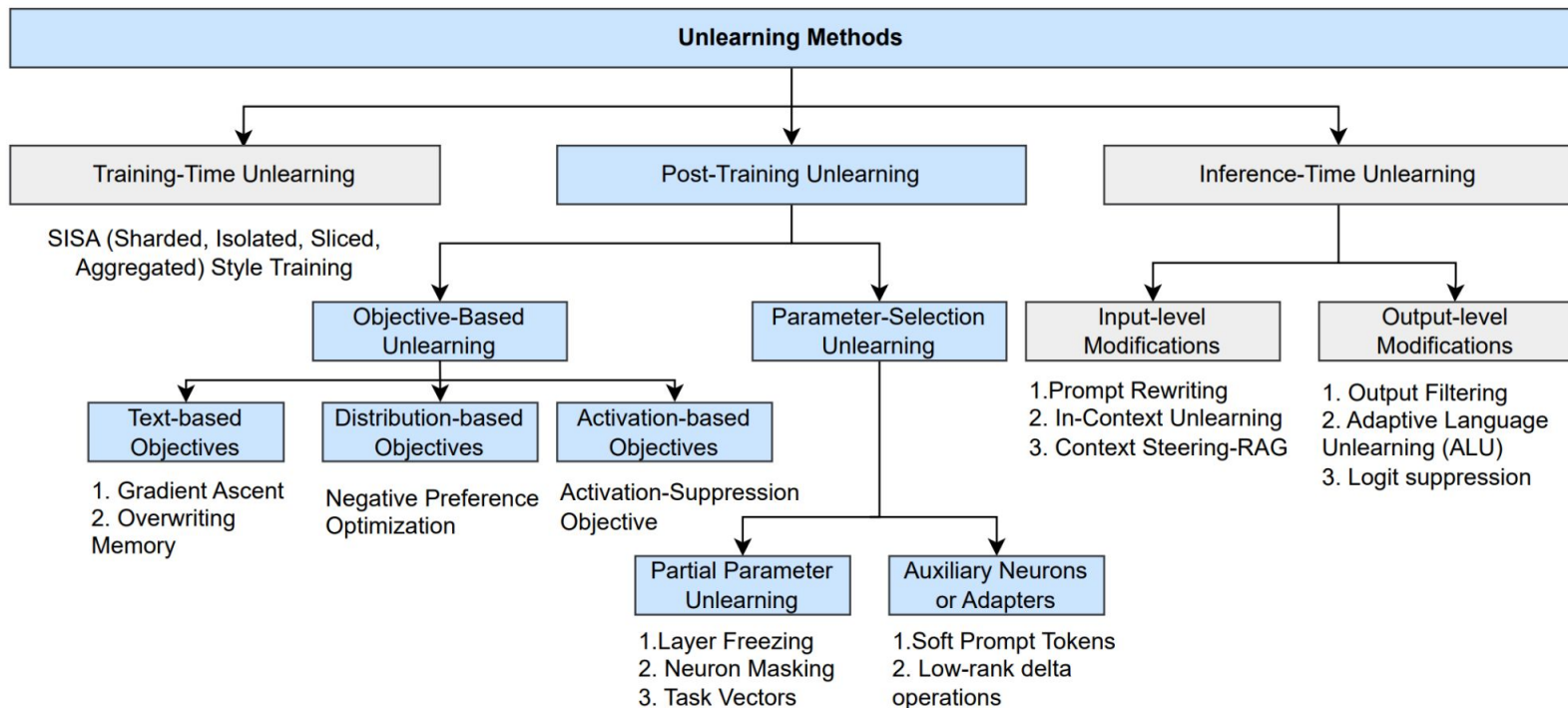
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- SISA Extensions for Modern LLMs
 - Adapter-based SISA (Store LoRA/adapters per shard → remove only small modules)
 - FairSISA (Ensures forgetting one demographic does not cause bias shifts)

Question: What problem do you see with SISA for LLM Unlearning?



[SISA Framework](#)

Taxonomy of Unlearning Methods in LLMs



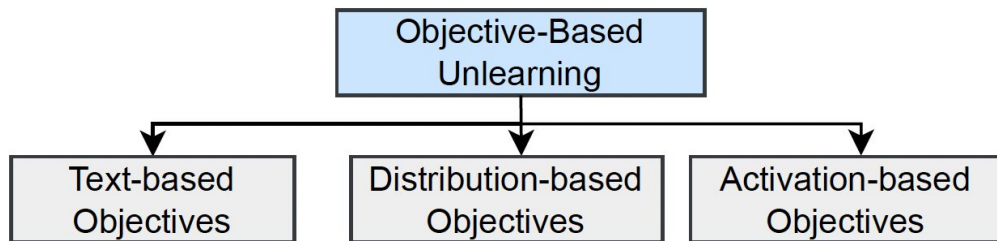
Post-Training Unlearning

- Modify an already-trained model to erase specific knowledge
- Most active area of research (works on deployed LLMs)
- Two design questions:
 - What objective?
 - Which parameters to update?



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Text-Based Objectives

- Operate directly on tokens / outputs
- Most common: gradient ascent on forget-set loss
- Negate cross-entropy for Df

$$L_{forget}(\theta) = -L_f(\theta)$$

- Substitution training: replace harmful facts with safe answers
- Token-level forgetting (WGA, FPGA) to avoid over-forgetting
- Risks:
 - Over-generalizing → degrade performance on related topics
 - Catastrophic forgetting spillover



Distribution-Based Objectives

- Match unlearned model's prediction distribution to a reference model
- Reference $\approx$ a model that “never learned” Df
 - Earlier checkpoint
 - Smaller model
 - A copy before fine-tuning
- **Core loss:**

$$L_{dist} = KL(p_{\theta}(y | x) || pref(y | x))$$

- Example: NPO (Negative Preference Optimization)
- More stable than raw gradient ascent
- Softer forgetting $\rightarrow$ less collateral damage



Activation-Based Objectives



- Target the hidden representations encoding D_f
- Identify activation signatures associated with forget data
- **Penalize or replace them:**

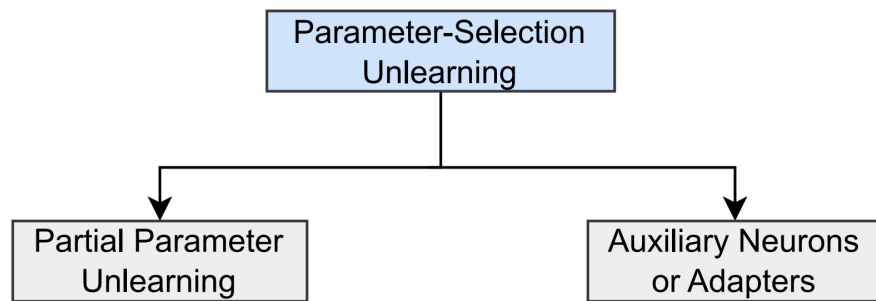
$$L_{act}(\theta) = \mathbb{E}_{x \in D_f} \| h\ell(x) - \mu_{r, \ell} \|_2$$

- Modify neuron/layer activations when seeing forget prompts
- Can remove deeper internal associations
- Harder: requires locating relevant neurons/directions

In practice, Unlearning objectives often combined (**Balance = key to preventing collapse**)

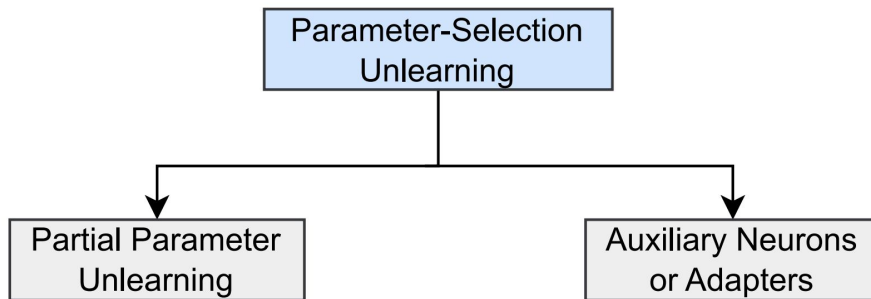
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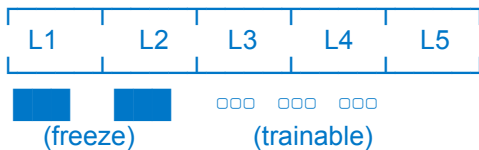
Question: Why full-model unlearning can be a problem?

Partial Parameter Unlearning (Targeted Edits)

- Only update layers/neurons influenced by Df
- Methods:
 - **Layer freezing** (train only top layers)



Model Layers



Layer Freezing

Partial Parameter Unlearning (Targeted Edits)

- Only update layers/neurons influenced by D_f
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 - **Neuron or attention-head masking**

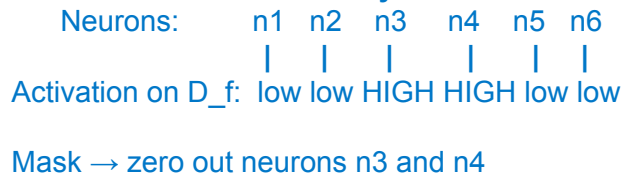


Model Layers



Layer Freezing

Layer L4



Neuron or Attention-Head Masking

Partial Parameter Unlearning (Targeted Edits)

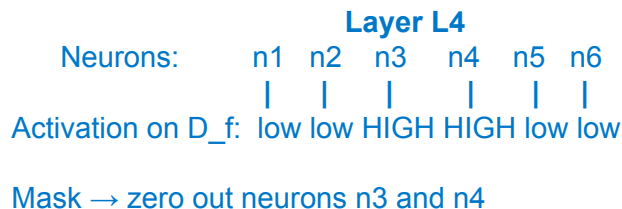
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 - **Task vector subtraction**



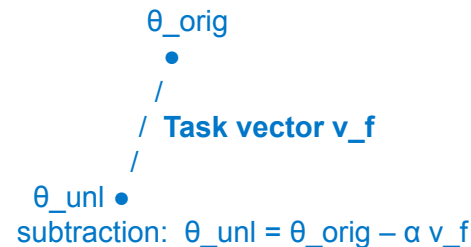
Model Layers



Layer Freezing



Neuron or Attention-Head Masking



Task Vector Subtraction

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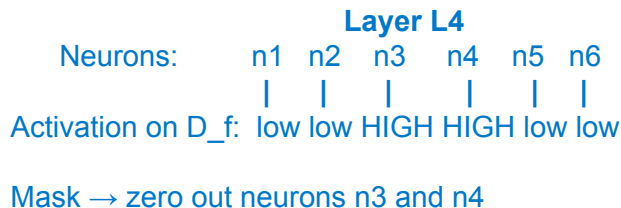


- Only update layers/neurons influenced by D_f
- Methods:
 - **Layer freezing** (train only top layers)
 - **Neuron or attention-head masking**
 - **Task vector subtraction**
 - **Influence-based selection**
- Goal: localize forgetting $\rightarrow$ reduce collateral damage
- Challenge: knowledge often distributed across many layers

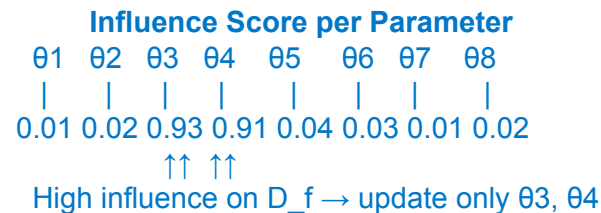
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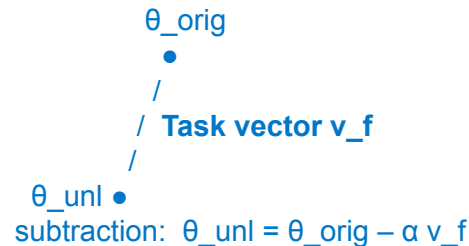
Layer Freezing



Neuron or Attention-Head Masking



Influence-Based Selection



Task Vector Subtraction

Soft Prompts & Adapters (Auxiliary Parameters)

- Add small, trainable modules to carry the influence of Df
- Techniques:
 - **Soft prompts / SPUL (2025)**
 - **LoRA-based unlearning adapters**
- Advantages: Main model weights remain unchanged, Lower compute + memory cost



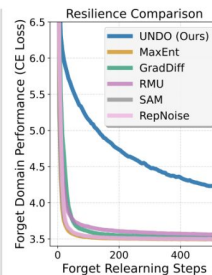
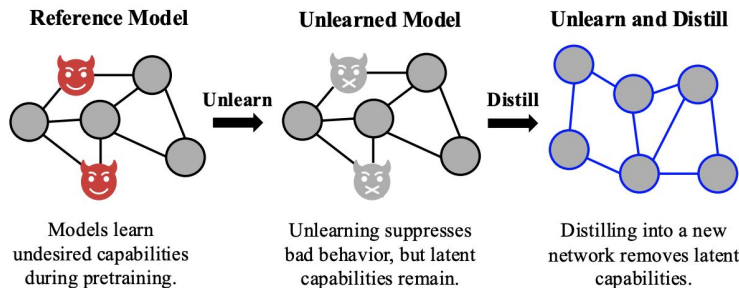
LoRA Unlearns More and Retains More

Model	UA		MIA-Efficacy		RA		TA		RTE	GPU
	5 Epochs	10 Epochs	5 Epochs	10 Epochs	5 Epochs	10 Epochs	5 Epochs	10 Epochs	(secs/epoch)	GB
ResNet-50										
Retrain	100.00	100.00	100.00	100.00	98.02	98.02	96.70	96.70	-	-
Finetune	100.00	100.00	100.00	100.00	92.52	96.27	88.29	91.44	137	6.9
Pruned	100.00	100.00	100.00	100.00	94.92	95.68	90.79	90.72	137	4.2
LoRA	97.22	100.00	100.00	100.00	96.90	97.19	95.03	93.49	122	5.8
Pruned LoRA	99.78	99.98	97.68	97.89	97.96	98.00	95.18	95.41	122	5.5
ViT										
Retrain	100.00	100.00	100.00	100.00	96.90	96.90	84.92	84.92	-	-
Finetune	100.00	100.00	100.00	100.00	87.64	86.80	79.79	79.94	132	3.2
Pruned	100.00	100.00	100.00	100.00	84.76	86.14	78.81	79.53	132	3.2
LoRA	87.66	95.58	98.14	99.58	97.72	97.81	85.33	85.16	48	0.7
Pruned LoRA	100.00	100.00	100.00	100.00	97.39	97.63	85.53	85.34	48	0.7

Table 1: Results of ResNet-50 and ViT When Tested Using Various Unlearning Approaches (in percent accuracy)

Auxiliary Models for Unlearning

- Train small helper models to counteract D_f
- Examples:
 - **Negative Distillation + Logit Difference (ULD)**
- Effect: cancel out forgotten knowledge without editing core weights



[Distillation robustifies unlearning](#)

[Reversing the Forget-Retain Objectives: An Efficient LLM Unlearning Framework from Logit Difference](#)

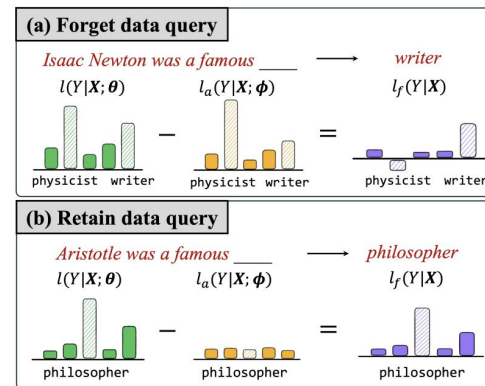
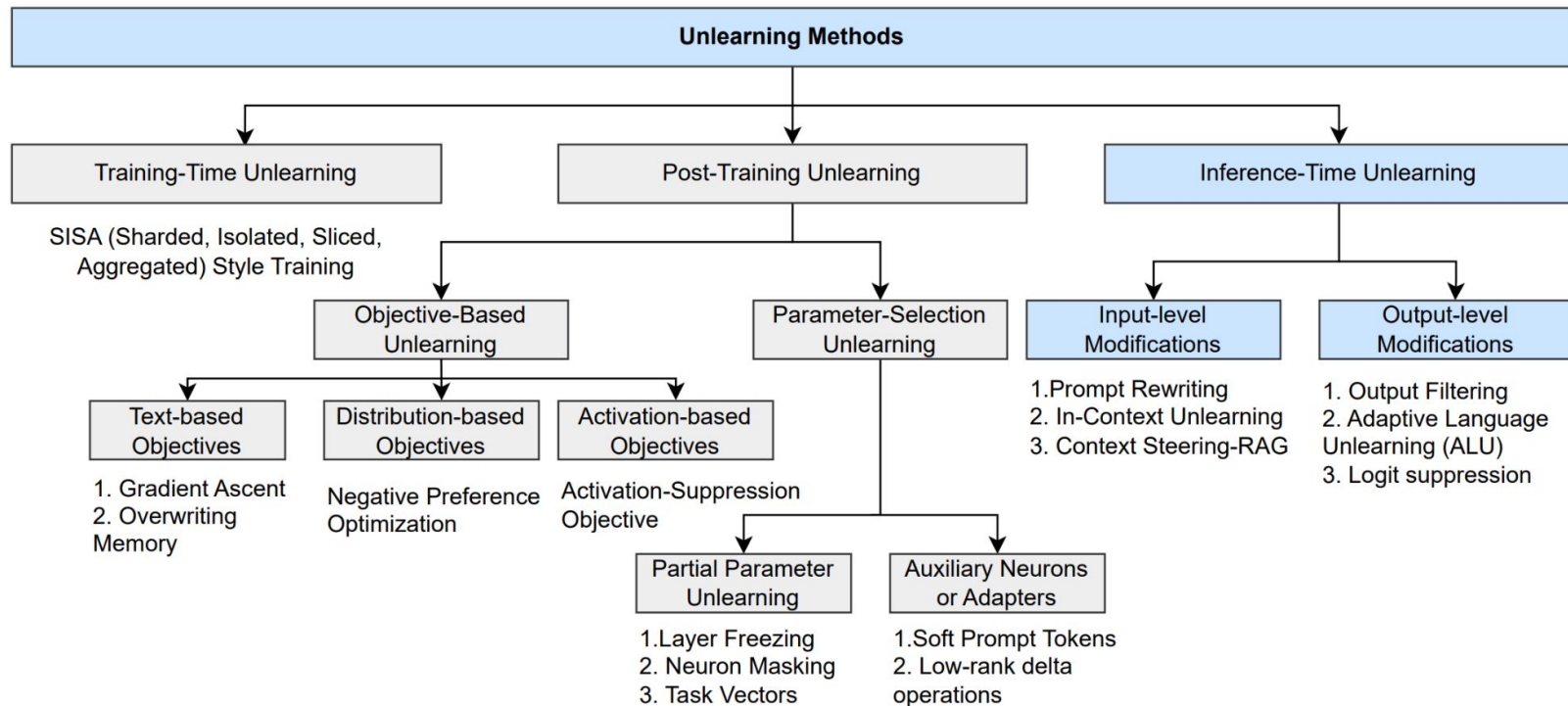


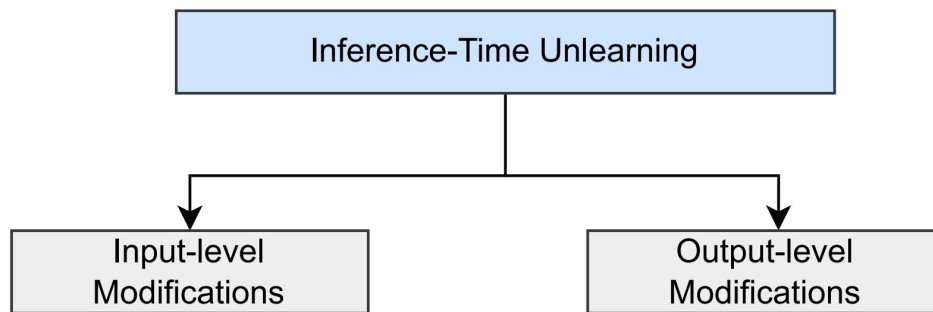
Figure 1: Illustration of the logit subtraction operation. We simulate the output distribution of an unlearned LLM using the assistant LLM's output.

Taxonomy of Unlearning Methods in LLMs



Inference-Time Unlearning

- No parameter updates
- Modify inputs or outputs in real time
- Practical, cheap, deployable immediately
- Acts as a wrapper around the model
- Often “pretend unlearning” rather than true forgetting



Input Level Modifications

- Change the model's input so it behaves as if it forgot.
- Techniques:
 - **Guardrail Prompting (System-Level Instructions)**

```
define flow politics
  user ask about politics
  bot inform cannot respond
```

```
define flow stock market
  user ask about stock market
  bot inform cannot respond
```

[Guardrail Prompting](#)



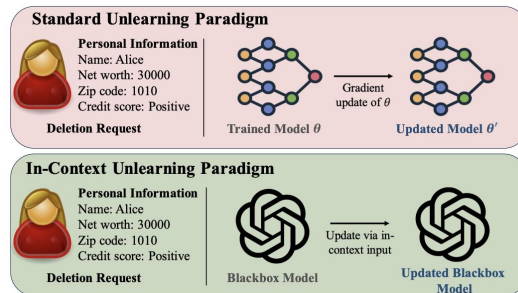
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 - **Input-Level: ICKU (Unlearning Tokens: <UNL> ... </UNL>)**



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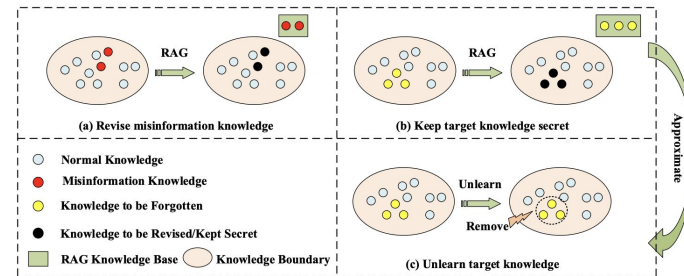
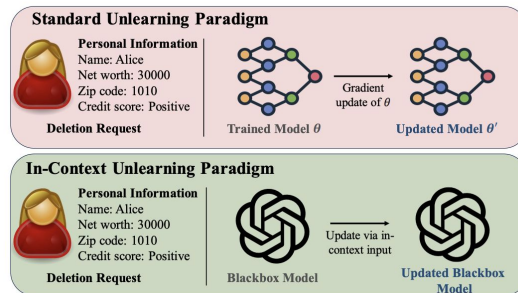
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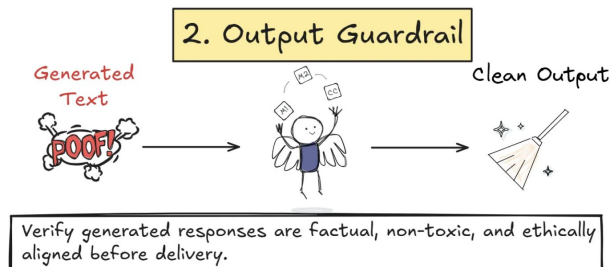
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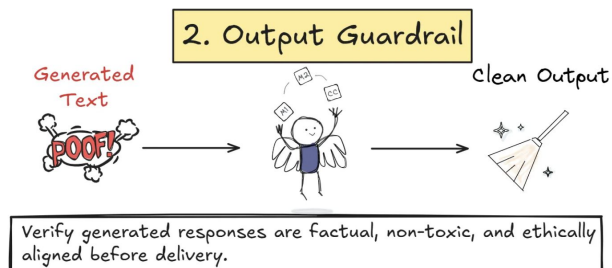
- Change the model's output so the forbidden content never reaches the user
- Techniques:
 - **Output Filtering**



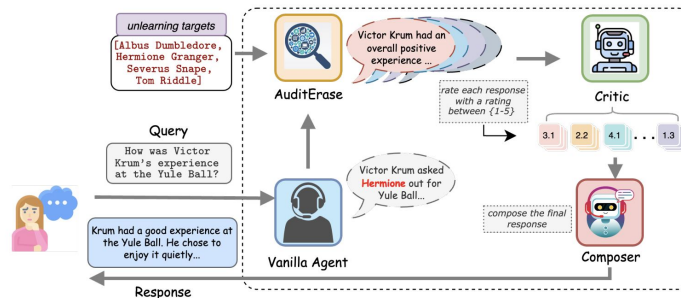
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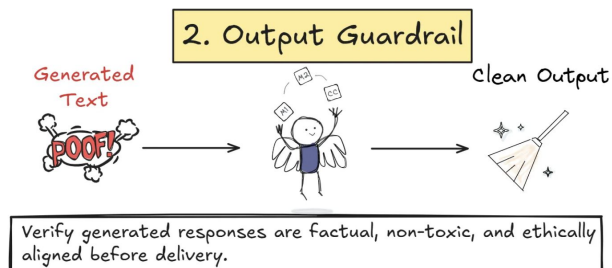
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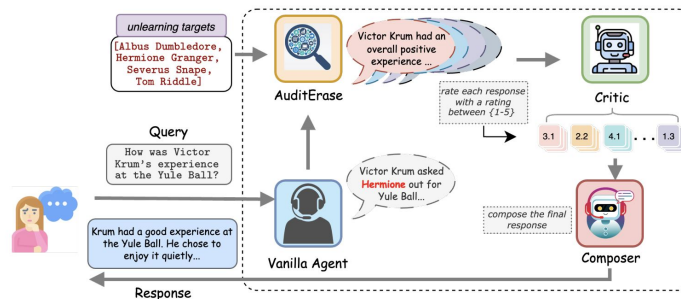
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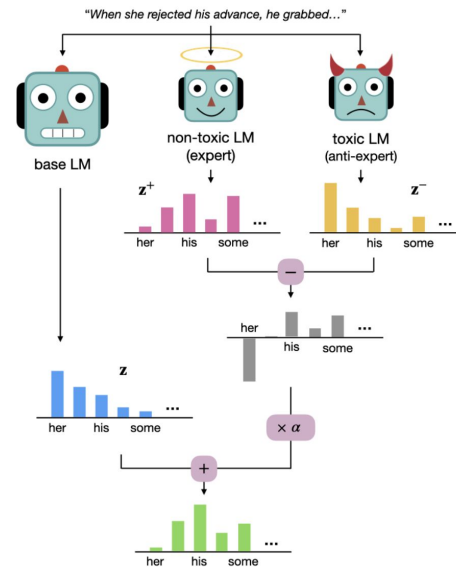
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 - **Multi-Agent Output Sanitization (ALU)**
 - **Logit Steering (DExperts, ULD-UNLEARNING)**



Output Filtering



Multi-Agent Output Sanitization



DExperts

Evaluating Unlearning Success

- Evaluating LLM Unlearning is non-trivial
 - Standard metrics (accuracy/perplexity) are not enough
- Evaluation must ask:
 - Did we forget D_f ?**
 - Did we preserve D_r ?**
 - How efficient & robust is the process?**
- Unlearning Success = 3-way trade-off

Forgetting Effectiveness ↔ Retained Utility ↔ Efficiency / Scalability



Forgetting Effectiveness



- **Direct evaluation on forget set Df**

- Accuracy drop on Df

$$AccDf \downarrow (\text{to chance / I don't know level})$$

- Likelihood / perplexity shift

$$\Delta PPLDf = PPLDf_{\text{after}} - PPLDf_{\text{before}} > 0$$

- Qualitative criteria: refusal / “I don’t know”

- **Membership Inference Attacks (MIA):** Detect if a sample was in training.

- If unlearning works (indistinguishable from unseen data):

$$MIA - AUC(Df) \rightarrow 0.5$$

- Pros: evidence of true removal
- Cons: costly, unstable in LLMs

- **Adversarial Probing:** Ensure no traces remain even under attack

- Paraphrases, Indirect probes, Prompt injection attempts, Multi-hop queries etc.

Retained Utility

Utility preservation checks

- Maintain accuracy / PPL on Dr

$$Acc_{Dr}^{after} \approx Acc_{Dr}^{before}$$

- General benchmarks (MMLU, MT-Bench, etc.)
- Spot “ripple effects” on related knowledge

Better method: Large drop on Df with minimal drop on Dr



Efficiency, Scalability, Sequential Requests

Efficiency metrics

- Time vs retraining:

$$Speedup = \frac{T_{retrain}}{T_{unlearn}}$$

- GPU / memory overhead
- Gradient steps needed

Scalability: Performance as $|D_f|$ grows

Sequential unlearning: Repeated requests shouldn't collapse utility



Standard Benchmarks

- **TOFU:** synthetic author/profile forgetting
- **RWKU:** real-world entity knowledge targets + MIA/adversarial suite
- **WMDP:** hazardous knowledge proxy for bio/cyber/chem security
- **MUSE:** six-way evaluation (privacy/utility/scalability/sequential etc.)



Challenges and Open Issues

Quantization-Induced Catastrophic Memory Recovery (Core Issue - [Paper](#))



- Unlearning often makes small weight changes (suppression, not removal)
- 4-bit quantization maps both the original and unlearned weights into the same quantization bucket
- This erases the forgetting update and restores the original representation (21% to 83% = 62%)

FULL-PRECISION WEIGHTS

Original Model
($w_{\text{orig}} = 0.152$)

Unlearned Model
($w_{\text{unlearn}} = 0.067$)

Bucket	Index	Value Range	Quantized Value
1111		0.20 – 0.30	0.25
1010		0.10 – 0.20	← BOTH map here → 0.15
0110		0.00 – 0.10	0.05
0010		-0.10 – 0.00	-0.05

Quantized Original Model

→ 0.15 (bucket 1010)

Quantized Unlearned Model

→ 0.15 (bucket 1010)

Other Open Challenges

- Ambiguous definitions: What counts as “forgotten”? refusal vs. ignorance vs. chance-level
- Boundary trade-off: hard to find λ that forgets D_f without collapsing D_r
- Identifying full scope: indirect, derived, or paraphrased data hard to trace
- Scaling to 70B+ models: methods break under large D_f or sequential requests
- Multilingual leakage: forgotten in English $\neq$ forgotten in Urdu/French
- Multimodal Unlearning: Forgetting a concept in one modality (e.g., text) does not erase it in others (images, audio, OCR), making cross-modal concept removal difficult
- Over-unlearning: ripple/onion effects wiping nearby knowledge
- Attack resilience: prompt injection, fine-tuning, adversarial probes still resurrect memory



